

CSA0706 Computer Networks

Assignment

Assignment Title	Design and Evaluate a Smart Campus Network Using Protocol Analysis, Network Topology and Simulation-Based Performance Metrics
Course Outcome(s) – CO	CO3: Analyze and compare the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions. CO4: Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards CO5: Measure network performance using key metrics with simulation tools
Bloom's Taxonomy Level	L5 – Evaluate; L6 – Create
SDG Mapping	SDG 3 – Industry, Innovation and Infrastructure

Problem Statement

Design and simulate a Smart University Campus Network connecting at least four network zones and 1000 end devices such as PCs, laptops, servers, IP cameras, wireless clients, and IoT devices. Develop and compare Star, Mesh, Bus, and Hybrid topologies using suitable networking devices and media.

Implement the selected topology using Cisco Packet Tracer, GNS3, EVE-NG, or an equivalent simulator. Analyze TCP, UDP, HTTP, DNS, FTP, and ICMP traffic under normal academic use, peak online-examination traffic, and congestion/link-failure conditions.

Measure key performance metrics such as throughput, delay, packet loss, jitter, response time, and bandwidth utilization. Present the results using tables and graphs, identify bottlenecks, compare the four topologies, and recommend the most suitable campus network architecture based on performance, reliability, scalability, cost, and resource utilization. Conduct repeated simulation trials where necessary to improve the reliability of the observations, record the collected data systematically, and represent the results using tables, charts, and comparative analysis.

Detailed Assessment Rubric with Performance Levels (Mapped to Assignment

Requirements)

Assessment Criterion	Marks	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
1. Problem Understanding & Network Requirements	10	Clearly identifies application needs, traffic patterns, topology constraints, device/media requirements, performance objectives and	Identifies most requirements with minor omissions.	Identifies basic requirements but lacks depth.	Requirements are incomplete or poorly understood.

		test conditions.			
2. Protocol Performance Analysis	20	Systematically analyzes TCP, UDP, HTTP,FTP,ICMP and DNS under multiple controlled conditions and explains differences using quantitative and qualitative evidence.	Provides a sound comparison with adequate evidence.	Provides a basic comparison with limited evidence.	Analysis is inaccurate, superficial, or unsupported.
3. Topology Design & Network Infrastructure	15	Creates technically correct star, mesh, bus and hybrid topologies and selects appropriate networking devices, media and cabling standards with clear justification.	Most topology and infrastructure choices are appropriate.	Basic designs are present but contain gaps or weak justification.	Designs are incomplete or technically unsuitable.
4. Simulation Implementation & Experiment Design	15	Builds the proposed network and controlled experiments in a suitable simulator; test cases are repeatable, clearly defined and technically valid.	Implementation and experiments are mostly correct.	Simulation is functional but experiments are limited or poorly controlled.	Simulation or experiment design is incomplete.

5. Network Performance Measurement	15	Accurately measures multiple metrics such as latency, throughput, packet loss, jitter, delay and/or delivery rate; results are organized and repeatable.	Measures appropriate metrics with adequate accuracy.	Measures a limited set of metrics or has weak measurement methodology.	Performance metrics are missing, inaccurate, or not reproducible.
6. Results Interpretation & Engineering Decisions	10	Uses tables/graphs to interpret results, identifies bottlenecks and trade-offs, and recommends topology/protocol choices based on measured evidence.	Interprets results and gives reasonable recommendations.	Provides limited interpretation or weak recommendations.	Conclusions are unsupported by measured results.
7. Broader Considerations & Professional Responsibility	5	Considers reliability, scalability, standards, efficient resource use, responsible experimentation, and practical	Addresses most broader considerations adequately.	Limited discussion of broader design or professional considerations.	Negligible consideration of broader impacts.

		network deployment considerations.			
8. Technical Documentation & Reflection	10	Provides a professional report with diagrams, methodology, configurations, raw/processed results, limitations, references and a thoughtful 300–500 word reflection.	Complete report and reflection with good clarity.	Documentation or reflection lacks depth.	Poor, incomplete, or missing documentation/reflection.

TOTAL
100

Submission Requirements

Students shall submit the following deliverables as part of the assignment evaluation:

S. No.	Deliverable	Description
--------	-------------	-------------

1	Technical Report	Comprehensive report covering problem definition, topology alternatives, selected architecture, protocol experiments, measurement methodology, results, analysis, engineering decisions, limitations and conclusions.
2	Network Topology Diagrams	Logical and physical diagrams for star, mesh, bus and hybrid designs showing devices, links, transmission media and cabling standards.
3	Simulation / Emulation Project Files	Original project files generated using Packet Tracer, GNS3, EVE-NG, or equivalent simulation/emulation platforms.
4	Experiment & Test Plan	Clearly defined test cases, traffic/load conditions, variables, measurement procedure, metric definitions and repetition strategy.
5	Protocol Performance Evidence	Packet captures, simulator outputs, logs or screenshots supporting TCP, UDP, HTTP, FTP, ICMP and DNS comparisons.
6	Performance Measurement Results	Raw/processed measurements for at least five key metrics, with units, test conditions, tables and graphs.
7	Engineering Analysis & Design Justification	Comparative analysis of protocol and topology performance, bottlenecks, trade-offs, limitations and justification of final design choices.
8	Simulation Configuration & Validation Evidence	Relevant configurations, connectivity verification, test

		outputs, screenshots and evidence demonstrating correct implementation and troubleshooting.
9	Individual Reflection (300-500 words per student)	Each student must submit an individual reflection addressing the most challenging aspect, most useful networking concept, technical problem encountered, how it was solved, lessons learned from simulation tools, an engineering decision they would change, and how the assignment improved understanding of computer networks.
10	Presentation / Demonstration	Demonstration of the alternative topologies, simulation experiments, protocol comparisons, performance measurements, graphs/results and final engineering recommendations.

Deliverables

To receive full credit, each submission must include the following deliverables:

1. Pseudo code
2. Implementation & Results
3. Github Upload

1. Problem Statement and Problem Formulation

Modern universities operate as always-on digital campuses: lecture delivery, research computing, physical security, hostel connectivity, and — critically — high-stakes online examinations all depend on a single shared network fabric. A design that performs adequately under normal academic load can fail precisely when it matters most: during a scheduled exam window when thousands of concurrent sessions hit the same servers within a narrow time band.

This project designs and simulates a campus network connecting at least four logically and physically distinct zones and approximately 1000 end devices (PCs, laptops, servers, IP cameras, wireless clients and IoT sensors). Four candidate topologies — Star, Mesh, Bus and Hybrid — are built and stress-tested under identical traffic conditions so that the trade-offs between them can be measured rather than assumed, and a final architecture recommended on evidence.

1.1 Campus Zones

The campus is modelled as five zones (satisfying the "at least four" requirement with one zone held in reserve to demonstrate scalability):

- Academic Block Zone — lecture-hall PCs, staff laptops, projector/AV endpoints, classroom Wi-Fi (≈350 devices)
- Administration Zone — ERP/finance workstations, HR systems, biometric/access-control IoT (≈100 devices)
- Library & Learning Commons Zone — public PCs, laptops via Wi-Fi, digital-resource kiosks (≈150 devices)
- Hostel / Residential Zone — student personal devices (laptops, phones) over Wi-Fi, wired warden offices (≈300 devices)
- Data Center / Server Farm Zone — application, DNS, FTP, web/exam-portal, authentication and monitoring servers, plus IP camera NVR (≈40 servers + 60 IP cameras across campus, backhauled here)

Total end devices: ≈1000, distributed roughly as above and finalised once the Packet Tracer/GNS3 project is built.

1.2 Objective and Expected Outcomes

- Design logical and physical topologies (Star, Mesh, Bus, Hybrid) for the same campus requirement set.
- Implement one primary architecture (Hybrid, selected in Section 3) end-to-end in a simulator, with the other three built at representative scale for comparative measurement.

- Generate and capture TCP, UDP, HTTP, DNS, FTP and ICMP traffic under three conditions: normal academic use, peak online-examination load, and congestion / link-failure.
- Measure throughput, end-to-end delay, packet loss, jitter, response time and bandwidth utilization for each topology under each condition.
- Identify bottlenecks and single points of failure, and recommend the most suitable architecture on performance, reliability, scalability, cost and resource-utilization grounds.

1.3 Requirements, Constraints and Assumptions

Functional requirements

- Any device in any zone must reach the Data Center (DNS, FTP, exam-portal/HTTP servers) and the campus edge/Internet.
- Inter-zone routing must support VLAN segmentation per zone (security and broadcast-domain isolation).
- Wireless coverage for classrooms, library, and hostel zones via access points on a dedicated management VLAN.
- IP camera and IoT traffic must be isolated on a separate VLAN with rate-limited access to the core.

Non-functional requirements

- Support ≈ 1000 concurrent/registered endpoints with headroom for growth (scalability).
- Tolerate at least one link or device failure in the core without total zone isolation (reliability/redundancy).
- Sustain acceptable response time (< 150 ms LAN, < 2 s HTTP page/exam-submission) during peak exam load.
- Remain deployable at reasonable capital/operational cost — avoid over-engineering every access link to core-grade redundancy.

Constraints

- Simulator device/interface count and performance limits (Packet Tracer typically handles hundreds, not 1000, live nodes fluently — addressed via sub-netted representative clusters, see Section 4).
- Fixed 2-day assignment timeline; simulation trials must be efficient and repeatable.

Assumptions

- Campus buildings are within standard structured-cabling distances (≤ 100 m copper per horizontal run); inter-building backbone uses fiber.
- Exam-peak traffic model: $\sim 70\%$ of hostel + academic devices open the exam portal within a 15-minute login window.
- Congestion/failure scenario simulated by saturating a core link and independently by administratively shutting down a core/distribution link.

1.4 Traffic and Application Profile

Application	Primary Protocol(s)	Typical Zones	Load Characteristic
Web / LMS / Exam Portal	HTTP, TCP	All zones $\rightarrow$ Data Center	Bursty; extreme peak during exams

Name Resolution	DNS, UDP	All zones → Data Center	High frequency, small packets
File / Course-Material Transfer	FTP, TCP	Academic, Library → Data Center	Steady, throughput-sensitive
Video Surveillance (IP Cameras)	UDP/RTP, TCP (NVR mgmt)	All zones → Data Center	Continuous, bandwidth-heavy
IoT Telemetry (access control, sensors)	UDP/MQTT-like, ICMP	Admin, Hostel	Low-rate, latency-tolerant
Network Diagnostics	ICMP	All zones	Low volume, used for fault detection
Voice/Video Conferencing (optional)	UDP, RTP	Academic, Admin	Jitter- and delay-sensitive

2. Topology Design & Network Infrastructure

All four topologies connect the same five zones and the same device population; only the interconnection pattern, redundancy and device roles change. A three-tier hierarchical model (Core – Distribution – Access) is used as the reference architecture, with topology variation applied primarily at the Core/Distribution level, since applying Bus or full Mesh at the 1000-device access layer would be technically unsuitable (per rubric: designs must be "technically correct").

2.1 Common Building Blocks

Device Role	Example Device	Function
Access Switch	Cisco Catalyst 2960 (L2, 24/48-port)	Connects end devices per zone/floor; VLAN assignment
Distribution Switch	Cisco Catalyst 3560/3650 (L3)	Inter-VLAN routing per zone; aggregates access switches
Core Switch/Router	Cisco Catalyst 3850 / ISR 4331	Backbone interconnection between zones and Data Center
Wireless AP + WLC	Cisco Aironet 2800 + WLC 2504	Wi-Fi for classrooms, library, hostel zones
Firewall / Edge Router	Cisco ASA 5506-X / ISR	Internet edge, DMZ for public-facing servers
Servers	Generic Server (DNS, FTP, HTTP, DHCP)	Core campus services in Data Center zone
IP Camera / IoT Node	Generic IP Camera, IoT sensor	Surveillance and building automation

Transmission media and cabling standards

- Access layer (end device → access switch): Cat6 UTP, 1000BASE-T, ≤ 90 m horizontal run (TIA/EIA-568 structured cabling).
- Distribution ↔ Access (intra-building): Cat6a UTP or multimode fiber (OM3/OM4), 1–10 Gbps.
- Core ↔ Distribution (inter-building backbone): Single-mode fiber (OS2), 10 Gbps, per TIA-568.3-D.
- Wireless access: IEEE 802.11ac/ax, WPA2/WPA3-Enterprise on a dedicated wireless VLAN.
- IoT / camera segment: Cat6 UTP or PoE+ (802.3at) links to access switches, isolated VLAN.

2.2 Star Topology

Every zone's access switches connect in a pure hub-and-spoke pattern to a single central core switch/router; every distribution switch also has exactly one uplink to the same core.

- Devices: 1 central core switch, 5 distribution switches (one per zone), access switches per building/floor beneath each.
- Media: Fiber uplinks from each distribution switch to the single core device.
- Strength: Simple to design, troubleshoot and cost-model; centralized management and policy enforcement.
- Weakness: The core switch is a single point of failure — its loss isolates all five zones simultaneously; core port count and backbone bandwidth must be over-provisioned for 1000 devices.

2.3 Mesh Topology

Distribution/core devices for each zone are interconnected with redundant links to multiple peers rather than one central node. A full mesh links every core-level node to every other; a partial mesh (used for the practical build) links each zone to at least two other zones/paths.

- Devices: 5 zone core/distribution routers meshed with 2–4 redundant fiber links each; dynamic routing (OSPF) for path selection and failover.
- Media: Multiple single-mode fiber backbone links per node, with link-state routing to exploit path diversity.
- Strength: High fault tolerance — no single link or node failure isolates a zone; multiple active paths can improve aggregate throughput.
- Weakness: Highest cabling and port cost (full mesh scales as $n(n-1)/2$ links); routing/configuration complexity increases significantly.

2.4 Bus Topology

Included for mandated comparison. All zone segments tap off a single shared backbone cable (logical bus, e.g., a shared coax or a hub-emulated shared-medium segment in the simulator) rather than switched point-to-point links.

- Devices: Legacy hubs/shared-medium segments per zone tapped onto one linear backbone; terminators at each end.
- Media: Coaxial (10BASE2) or a hub-based shared Ethernet segment emulated in the simulator, since pure bus media is obsolete for structured campus builds.
- Strength: Lowest cabling cost for a small, non-critical segment; trivial to explain and diagram.
- Weakness: A single backbone break disconnects every zone beyond the break; shared medium means collisions/contention scale badly well before 1000 devices — technically unsuitable as a primary

campus backbone, which is precisely the conclusion the comparative data in Section 6 is expected to support.

2.5 Hybrid Topology (Proposed Primary Architecture)

A hierarchical Core–Distribution–Access design: partial-mesh redundant fiber at the Core (between the Data Center and each zone's distribution router), star connections from each distribution switch down to its access switches, and star/AP connections from access switches to end devices. This is the architecture carried forward into simulation in Section 4.

- Core: Partial mesh (each zone has ≥ 2 physically diverse fiber paths to the Data Center core) — inherits Mesh's fault tolerance where it matters most.
- Distribution → Access: Star — inherits Star's simplicity and manageability where full redundancy is not cost-justified.
- Access → End devices: Star (wired) / AP-based (wireless) — standard, scalable, easy to troubleshoot per device.
- Redundancy devices: HSRP/VRRP pair of core routers at the Data Center; EtherChannel on distribution uplinks.

2.6 Topology Comparison (Design-Level)

Criterion	Star	Mesh	Bus	Hybrid
Fault tolerance	Low (single core = SPOF)	High (multi-path)	Very Low (backbone break)	High (redundant core only)
Cabling / device cost	Low–Medium	High	Very Low	Medium
Scalability to 1000+ devices	Medium (core bottleneck)	High but costly	Poor	High
Manageability / troubleshooting	Easy	Complex	Easy but fragile	Moderate
Expected throughput under load	Medium	High	Low	High
Suitability as campus backbone	Acceptable, small campus	Ideal but expensive	Not suitable	Recommended

Section 6 (Results Interpretation) of the full report will confirm or revise this design-level ranking using measured throughput, delay, jitter, loss and utilization data from Section 5, per topology per traffic condition.

3. Experiment & Test Plan

3.1 Representing 1000 Devices at Simulator Scale

Packet Tracer/GNS3 remain responsive with a few hundred fully modelled nodes; 1000 devices are represented using a scaled cluster technique so results remain valid and repeatable:

- Each zone builds one fully modelled access switch stack (e.g., 40–60 real end devices with real IP/MAC config).
- Remaining devices per zone are represented as traffic load using the simulator's traffic generator / scripted ping-HTTP-FTP jobs against the servers, scaled proportionally (documented scaling factor, e.g., 1 simulated session = 15 real users) so throughput/utilization figures can be extrapolated to the full 1000-device population.
- This scaling factor and its justification is recorded explicitly in the report per rubric requirement for "repeatable, clearly defined, technically valid" experiments.

3.2 Test Conditions

Condition	Description	Traffic Generated
Normal Academic Use	Baseline weekday load: browsing, LMS access, occasional file downloads, routine camera feed, background IoT telemetry, periodic DNS lookups	HTTP, DNS, FTP (light), ICMP (periodic), UDP (camera/IoT, steady low rate)
Peak Online-Examination Traffic	~70% of Academic + Hostel devices access exam portal within a 15-minute login window; simultaneous DNS bursts, TCP session floods to exam server, minimal FTP/camera unaffected	HTTP/TCP (spike), DNS (spike), ICMP (monitoring), UDP (steady camera baseline)
Congestion / Link-Failure	(a) Saturate one core/backbone link with generated background traffic to observe congestion behaviour; (b) administratively shut down one core or distribution link/device to observe failover and isolation impact	All protocols continued at normal-use levels while congestion/failure is induced

3.3 Metrics, Definitions and Collection Method

Metric	Definition	Collection Method	Unit
Throughput	Successfully delivered data volume per unit time on a link/server	Simulator link statistics; Wireshark I/O graph on captured pcap	Mbps
End-to-end delay	Time from packet transmission to reception at destination	Ping RTT/2 for ICMP; PDU inspection in simulator for other protocols	ms
Packet loss	Percentage of packets sent but not received	Ping loss %; simulator PDU list (sent vs. received count)	%
Jitter	Variation in inter-packet delay, most relevant to UDP/camera streams	Wireshark RTP/UDP stream statistics	ms

Response time	Time from client request to completed HTTP/FTP transaction	Simulator HTTP client stats; Wireshark TCP stream time	s
Bandwidth utilization	Used capacity as a percentage of link capacity	Simulator link load indicator / interface counters	%

3.4 Repetition and Reliability Strategy

- Each (Topology × Condition) combination is run a minimum of 5 trials; mean and standard deviation are reported, not single-run figures.
- Trials are spaced to avoid simulator-state carryover (device reload / topology reset between runs).
- Outlier trials (e.g., simulator glitch, misconfiguration) are flagged, excluded with justification, and re-run.
- A fixed random seed / identical background-traffic script is reused across topologies so that only the topology variable changes between comparable runs — controlling for confounding factors.

3.5 Test Case Matrix

4 topologies × 3 conditions × 6 metrics × 5 trials = 360 measurement records, organised into one master results spreadsheet (raw) and one summary spreadsheet (mean/SD per cell) for graphing in Section 6.

4. Pseudocode / Experiment Automation Logic

The following pseudocode describes the controlled-experiment procedure used to drive traffic generation, timing, and metric logging consistently across all four topologies (implemented via Packet Tracer scripting / scripted PDU generation and companion Wireshark captures, or equivalent Python + simulator API on GNS3/EVE-NG).

4.1 Master Experiment Driver

```

FOR EACH topology IN [Star, Mesh, Bus, Hybrid]:
  LOAD topology_project_file(topology)
  FOR EACH condition IN [Normal, ExamPeak, CongestionFailure]:
    FOR trial = 1 TO 5:
      RESET_DEVICE_STATE(topology)
      START_CAPTURE(all_core_links)
      GENERATE_TRAFFIC(condition)           // see 4.2
      WAIT(observation_window)
      RECORD = COLLECT_METRICS()           // see 4.3
      STOP_CAPTURE()
      APPEND_TO_RESULTS_LOG(topology, condition, trial, RECORD)
    END FOR
  END FOR
END FOR

```

```
END FOR
```

```
COMPUTE_MEAN_AND_STDDEV(results_log)    // per topology-condition-metric cell  
EXPORT_TABLES_AND_GRAPHS(results_log)
```

4.2 GENERATE_TRAFFIC(condition)

```
IF condition == Normal:
```

```
    SEND periodic ICMP pings (all zones -> core, every 30s)  
    SEND HTTP requests (light rate, all zones -> web/LMS server)  
    SEND DNS queries (moderate rate, all zones -> DNS server)  
    SEND FTP transfer jobs (occasional, Academic+Library -> FTP server)  
    STREAM UDP camera feed (steady low bitrate, all zones -> NVR)
```

```
ELSE IF condition == ExamPeak:
```

```
    SELECT 70% of Academic + Hostel devices  
    WITHIN 15-minute window: EACH selected device opens TCP/HTTP session to  
exam server  
    EACH session issues a DNS lookup before connecting  
    CONTINUE baseline ICMP + UDP camera traffic unchanged (control variable)
```

```
ELSE IF condition == CongestionFailure:
```

```
    RUN Normal traffic generation (baseline)  
    BRANCH (a) Congestion: INJECT high-rate UDP flood on one core/backbone link  
    BRANCH (b) Failure: ADMIN-DOWN one core or distribution link/device at t =  
observation_window / 2  
    LOG time-to-reconverge (Mesh/Hybrid) or time-to-total-isolation (Star/Bus)  
END IF
```

4.3 COLLECT_METRICS()

```
throughput      = BYTES_DELIVERED(link) / observation_window  
delay           = AVERAGE(ping_RTT) / 2  
packet_loss_pct = (packets_sent - packets_received) / packets_sent * 100  
jitter          = STDDEV(inter_packet_delay) FOR udp_stream  
response_time   = timestamp(HTTP_response) - timestamp(HTTP_request)  
bandwidth_util  = link_load / link_capacity * 100  
RETURN {throughput, delay, packet_loss_pct, jitter, response_time,  
bandwidth_util}
```

5. Next Steps (Not Yet Covered in This Draft)

- Build the four .pkt/.gns3 project files per Section 2 designs and Section 3 scaling method.
- Execute the test matrix in Section 3.5; export raw logs into the results spreadsheet.
- Produce Section 5 (Performance Measurement tables/graphs), Section 6 (bottleneck identification, topology recommendation with evidence), Section 7 (reliability/scalability/SDG/broader considerations) and the individual reflections.
- Capture configuration screenshots, connectivity verification (ping/traceroute), and Wireshark evidence per protocol for the Protocol Performance Evidence deliverable.
- Push project files, report and README to GitHub; link in Google Classroom submission.